

Q1: A region R in the xy -plane is given. Find equations for the transformation T that maps a rectangular region S in the uv -plane onto R , where the sides of S are parallel to the u - and v -axes.

R is bounded by the hyperbolas $y = 1/x$, $y = 4/x$, and the lines $y = x$, $y = 4x$ in the first quadrant.

Find $T: (u, v) \mapsto (x, y)$

Hyperbola $y = 1/x$ $y = 4/x$. Line $y = x$ $y = 4x$

Let $u = xy$ and $v = y/x$, then we can see

$$\begin{aligned} y = 1/x &\Rightarrow u = 1 \text{ and } y = 4/x \Rightarrow u = 4 \\ y = x &\Rightarrow v = 1 \text{ and } y = 4x \Rightarrow v = 4 \end{aligned}$$

So use the transformation $u = xy$ and $v = y/x$ is good as it transforms our messy region in x, y into rectangular regions in u, v .

To find the transformation,

$$u = xy = x(vx) = vx^2 \Rightarrow x = \sqrt{u/v}$$

$$y = vx = v\sqrt{u/v} = \sqrt{uv}$$

So the region in S is the rectangle $1 \leq u \leq 4$
 $1 \leq v \leq 4$

Q2: Use the given transformation to evaluate the integral.

$$\iint_R (4x+8y) dA, \quad R \text{ is the parallelogram with vertices } (-1,3), (1,-3), (3,-1),$$

$$\text{and } (1,5); \quad x = \frac{1}{4}(u+v) \text{ and } y = \frac{1}{4}(v-3u)$$

1. Jacobian

$$x = \frac{1}{4}(u+v) \Rightarrow x_u = \frac{1}{4}, \quad x_v = \frac{1}{4}$$

$$y = \frac{1}{4}(v-3u) \Rightarrow y_u = -\frac{3}{4}, \quad y_v = \frac{1}{4}$$

$$J = \begin{vmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{vmatrix} = \begin{vmatrix} 1/4 & 1/4 \\ -3/4 & 1/4 \end{vmatrix} = 1/4$$

2. uv -Region

$$\begin{cases} x = \frac{1}{4}(u+v) \Rightarrow u+v = 4x \\ y = \frac{1}{4}(v-3u) \Rightarrow v-3u = 4y \end{cases}$$

$$\text{So } u+v = 4x \Rightarrow v = 4x - u$$

$$\text{Plug into } (4x - u) - 3u = 4y \Rightarrow u = x - y$$

$$\text{Then } v = 4x - u = 4x - (x - y) = 3x + y$$

$$\text{Hence } \begin{cases} u = x - y \\ v = 3x + y \end{cases}$$

$$\text{At } (-1,3), \quad u = -4 \quad v = 0$$

$$(1,-3), \quad u = 4 \quad v = 0$$

$$(3,-1), \quad u = 4 \quad v = 8$$

$$(1,5), \quad u = -4 \quad v = 8$$

So S is the rectangle $-4 \leq u \leq 4 \quad 0 \leq v \leq 8$

3. Function Transform

$$f(x,y) = 4x + 8y = 4\left(\frac{1}{4}(u+v)\right) + 8\left(\frac{1}{4}(v-3u)\right) = -5u + 3v$$

$$4. \quad \iint_R (4x+8y) dA = \iint_S (-5u+3v) \cdot \frac{1}{4} du dv = \frac{1}{4} \int_0^8 \int_{-4}^4 (-5u+3v) du dv$$

$$= \frac{1}{4} \int_0^8 \left[-\frac{5}{2} u^2 + 3uv \right] \Big|_{-\frac{4}{3}u}^{\frac{4}{3}u} dv$$

$$= \frac{1}{4} \int_0^8 -40 + 24v dv$$

$$= \frac{1}{4} \left[-40v + 12v^2 \right] \Big|_0^8$$

$$= 112$$

Q5: Evaluate the integral by making the appropriate change of variables.

(a) $\iint_R \frac{x-2y}{3x-y} dA$, where R is the parallelogram enclosed by the lines $x-2y=0$, $x-2y=4$, $3x-y=1$, and $3x-y=8$.

(b) $\iint_R \sin(9x^2+4y^2) dA$, where R is the region in the first quadrant bounded by the ellipse $9x^2+4y^2=1$.

a) ① Change of Variables

Note our bounds have form $\begin{cases} x-2y = \text{constant} \\ 3x-y = \text{constant} \end{cases}$

So define $\begin{cases} x-2y = u \\ 3x-y = v \end{cases}$

Then our boundaries become

$$\begin{cases} u=0 \\ u=4 \end{cases} \quad \begin{cases} v=1 \\ v=8 \end{cases}$$

$$\text{ie. } 0 \leq u \leq 4 \quad 1 \leq v \leq 8$$

② Solve x, y in terms of u, v

$$\begin{cases} u = x-2y \\ v = 3x-y \end{cases} \rightarrow \begin{cases} 3u = 3x-6y \\ v = 3x-y \end{cases}$$

Subtract
$$\begin{array}{r} 3u = 3x - 6y \\ v = 3x - y \\ \hline \end{array}$$

$$3u - v = -5y \Rightarrow y = \frac{v - 3u}{5}$$

Solve for x :

$$u = x - 2\left(\frac{v - 3u}{5}\right) = x - \frac{2v}{5} + \frac{6u}{5}$$

$$\Rightarrow x = u + \frac{2v}{5} + \frac{6u}{5} = \frac{5u + 2v + 6u}{5} = \frac{2v + 11u}{5}$$

Hence, $x = \frac{2v + 11u}{5}$, $y = \frac{v - 3u}{5}$

③ Jacobian

$$J = \begin{vmatrix} \partial x / \partial u & \partial x / \partial v \\ \partial y / \partial u & \partial y / \partial v \end{vmatrix}$$

$$= \begin{vmatrix} -1/5 & 2/5 \\ -3/5 & 1/5 \end{vmatrix} = 1/5$$

④ Function Transform

$$f(x, y) = \frac{x - 2y}{3x - y} = \frac{u}{v}$$

$$\begin{aligned} \textcircled{5} \iint_R \frac{x - 2y}{3x - y} dA &= \iint_S \frac{u}{v} |J| du dv \\ &= \int_0^4 \int_1^8 \frac{u}{v} \cdot \frac{1}{5} dv du \end{aligned}$$

Q6: Find the gradient vector field of f .

(a) $f(x, y) = y \sin(xy)$

(b) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$

(c) $f(s, t) = \sqrt{2s + 3t}$

(d) $f(x, y, z) = x^2 y e^{y/z}$

a) Recall gradient vector field of $f(x, y)$ is

$$\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$$

$$f(x, y) = y \sin(xy) \Rightarrow \frac{\partial f}{\partial x} = y \cos(xy) \cdot y = y^2 \cos(xy)$$

$$\Rightarrow \frac{\partial f}{\partial y} = \sin(xy) + xy \cos(xy)$$

$$\text{Hence, } \nabla f = \langle y^2 \cos xy, \sin(xy) + xy \cos(xy) \rangle$$

c) $\nabla f = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$

$$= \left\langle \frac{1}{\sqrt{2s+3t}}, \frac{1}{2\sqrt{2s+3t}} \right\rangle$$